Inverse Transition Learning from Expert Demonstrations:
Reproduction and a Joint Reward–Dynamics Extension

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1 Question

In standard reinforcement learning we know the transition dynamics T of a Markov decision process and want to compute an optimal policy π^*. Many real settings invert that assumption. In an intensive care unit, for example, the reward (patient outcome) is observable but the dynamics of how a patient responds to a particular dose are not. What we observe is a batch of clinician decisions that are close to optimal but not exactly optimal. The question this semester has been: given the reward R and a batch of ε-optimal expert transitions $D = \{(s, a, s')\}$, can we recover the dynamics T^* in a way that respects the optimality information in the actions, rather than throwing it away as a maximum-likelihood estimate over counts does?

The paper that frames this problem is *Inverse Transition Learning* [1]. The method I have been studying and reproducing is the convex quadratic program in their Algorithm 1.

2 Method and tools

An action a is ε-optimal at state s under dynamics T when $\max_{a'} Q^*(s, a'; T) - Q^*(s, a; T) \leq \varepsilon$. Observing the expert choose a at s therefore yields an inequality on Q^*. Writing the policy-evaluation inverse $v^\pi = (I - \gamma P^\pi)^{-1} r^\pi$ and substituting the Laplace-smoothed MLE T^{MLE} inside the inverse *linearizes* the constraint in T. The resulting program is

$$\min_T \ \tfrac{1}{2} \sum_{s,a,s'} N_{s,a,s'} \big(T_{s,a,s'} - T^{\mathrm{MLE}}_{s,a,s'}\big)^2 \quad \text{s.t.} \quad \varepsilon\text{-ball constraints on } (s,a) \in D, \text{ simplex on } T.$$

The solver alternates between a policy estimate $\hat{\pi}$ (uniform over observed actions at observed states; uniform over actions elsewhere, per Algorithm 1, line 3) and the QP above, terminating when the ε-ball property holds at every observed state. The convex pieces are standard tabular RL machinery from [2, Ch. 10]: Bellman optimality, policy-induced matrices, and the linear policy-evaluation system. The Laplace-smoothed MLE corresponds to a Dirichlet posterior mean from the same textbook's Chapter 11 on Bayesian model-based RL. The implementation uses CVXPY [5] with an OSQP–SCS–CLARABEL fallback chain.

3 Simulation: coverage sweep on Gridworld

I reproduced the paper's central empirical claim (their Figure 2) on a 5×5 Gridworld with soft walls at five tiles (penalty -5), slip probability 0.2, $\gamma = 0.95$, and $\varepsilon = 5.0$. The expert was 40% stochastic, meaning at 40% of states the expert mixes uniformly over the ε-ball rather than acting greedily. *Coverage* is the fraction of states that appear in D; for each observed (s, a) I sampled 10 transitions. I compared two estimators of T: the Laplace-smoothed MLE and ITL.

At full coverage, ITL achieves Normalized Value 0.998 ± 0.001 versus MLE's 0.142 ± 0.002, with 0.00 versus 7.20 constraint violations and ε-matching 1.000 versus 0.712 (averages over 5 seeds). MLE's argmax under-counts visited next-state cells when the expert is occasionally stochastic, so the greedy policy under $\hat{T}_{\mathrm{MLE}}$ walks into soft walls. ITL avoids this by enforcing that the expert's observed choices remain ε-optimal under $\hat{T}$. The qualitative shape matches the paper's Figure 2; the absolute scale runs roughly 1.5× larger than Table 4, almost certainly because the published soft-wall layout is not reported in detail, so my environment is somewhat less aversive. The structural pattern (ITL $\geq$ MLE on every

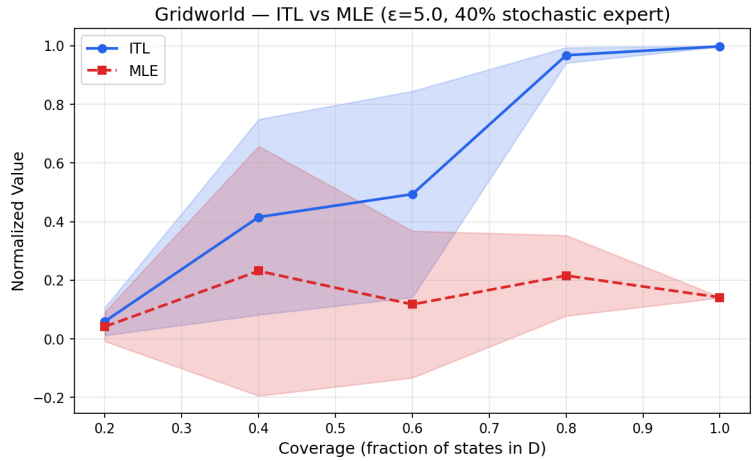

Figure 1: Normalized Value $V^{\pi^*(\hat{T})}(s_0; T^*)/V^{\pi^*(T^*)}(s_0; T^*)$ vs. coverage on Gridworld. ITL plateaus near 1.0 at full coverage and dominates MLE everywhere. Shaded bands: ± 1 std. across seeds.

metric, ε-matching $\to 1.0$ for ITL at full coverage as predicted by Theorem 1) holds on both Gridworld and a separate RandomWorld benchmark.

4 Open question: joint reward and dynamics recovery

The paper assumes R is known. In MIMIC-IV that assumption is fragile: the survival proxy reward (± 15 for discharge or mortality, after [4]) is hand-engineered. The thesis extension I have been prototyping, which the paper itself names as future work, is to lift R into a decision variable parameterized as $R(s, a) = \Phi(s, a)^\top w$ and solve for (T, w) jointly. The ε-ball constraint becomes

$$\big(\Phi(s, a) - \Phi(s, a')\big)^\top w + \gamma \left(T_{s,a} - T_{s,a'}\right)^\top v_{\text{lin}} \;\geq\; \varepsilon.$$

This is bilinear in (T, w) once v_{lin} is treated as a function of w, but each step of an alternation (hold w, solve for T; hold T, solve for w) is a convex QP. An L1 penalty on w plus an anchor at one (s, a) pin the additive ambiguity. Inspired by maximum-entropy IRL [3], but distinct from it because the dynamics are not assumed known.

A smoke test on the 3-state corridor with state-only one-hot features and the goal reward as anchor terminates at a fixed point in two outer iterations. The recovered policy matches the true optimal policy exactly. Transition MSE is 0.233, comparable to ITL alone with known R (0.237) and below the MLE baseline (0.282). The reward weights are recovered up to the L1 sparsity that the regularizer enforces on the small near-zero true weights, which is the expected identifiability behavior of one-hot state-only features. Stress tests on Gridworld and a baseline against MaxEnt-IRL with a separately estimated T are the next steps before the May 31 thesis draft.

Code and reproducibility. The full reproduction codebase (Python 3.10, CVXPY, MIT-licensed) is at `https://github.com/REPLACE-ME/itl-reproduction`, with inline cross-references between paper equations, textbook chapters, and source lines. A Colab that re-runs the corridor sanity check, the Gridworld coverage sweep at a reduced seed count, and the ITL+IRL smoke test that produced the numbers in Section 4 is at `https://colab.research.google.com/REPLACE-ME`.

[1] Benac, L., et al. (2024). Inverse Transition Learning: Learning Dynamics from Demonstrations. *AAAI 2025* (preprint).

[2] Krause, A., and Hübotter, J. (2025). *Probabilistic Artificial Intelligence*. Cambridge Univ. Press. Ch. 10–12.

[3] Ziebart, B. D., Maas, A., Bagnell, J. A., and Dey, A. K. (2008). Maximum Entropy Inverse Reinforcement Learning. *Proc. AAAI*.

[4] Komorowski, M., Celi, L. A., Badawi, O., Gordon, A. C., and Faisal, A. A. (2018). The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis in intensive care. *Nature Medicine* 24, 1716–1720.

[5] Diamond, S., and Boyd, S. (2016). CVXPY: A Python-Embedded Modeling Language for Convex Optimization. *JMLR* 17(83), 1–5.